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RANSOMWARE DETECTION AND PREVENTION SYSTEM

A CAPSTONE

PROJECT REPORT

Submitted in the partial fulfilment for the Course of

**CSA0361- Design and Analysis of Algorithms for Optimization
Technique**

to the award of the degree of
BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE ENGINEERING

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DECLARATION

We hereby declare that the report entitled “RANSOMWARE DETECTION AND PREVENTION SYSTEM” is a unique and original work submitted by us for the degree of **Bachelor of Engineering**. This report is the record of our capstone project for the course **CSA0515-Cryptography and Network Security**, carried out under the guidance of **Dr. T. Sangeetha**. We further declare that this work has not formed the basis for the award of any degree or diploma in this or any other University or institution of higher learning.

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BONAFIDE CERTIFICATE

Certified that this capstone project reports on “RANSOMWARE DETECTION AND PREVENTION SYSTEM” is the Bonafide work of V.Maduri Priya,G.Gnana Deepthi,Akshitha who carried out the capstone project work under my supervision for the course **CSA0515- Cryptography and Network Security**

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Submitted for the Project work Viva-Voce held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

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Sincerely,

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ABSTRACT

Ransomware is a rapidly growing cybersecurity threat that targets individuals, organizations, and institutions by encrypting important files and demanding a ransom for their recovery. Traditional antivirus systems mainly depend on known malware signatures, making them less effective against new and modified ransomware variants. To address this problem, the proposed **Ransomware Detection and Prevention System** is designed to detect suspicious activities at an early stage and prevent ransomware from causing severe damage to systems and data.

The proposed system continuously monitors system behavior, file activities, processes, and network-related events to identify patterns associated with ransomware attacks. It analyzes suspicious activities such as rapid modification or encryption of multiple files, creation of unusual file extensions, unauthorized process execution, abnormal file access, and sudden changes in system resources. A behaviour-based detection mechanism is used to identify both known and previously unseen ransomware threats. When suspicious behaviour exceeds a predefined threshold, the system generates an immediate security alert and initiates preventive measures.

The prevention component can terminate or block suspicious processes, restrict unauthorized file operations, isolate the affected system, and protect critical files from further modification. The system can also maintain secure backups of important data so that files can be restored without relying on ransom payments. A monitoring dashboard can provide information about detected threats, affected files, process activity, threat severity, and prevention actions taken by the system.

The major objective of this project is to provide a proactive and automated security solution that reduces the impact of ransomware attacks. Unlike approaches that respond only after files have been encrypted, the proposed system focuses on detecting abnormal behavior during the early stages of an attack. This helps minimize data loss, system downtime, financial damage, and recovery time. The system can be further enhanced by incorporating machine learning techniques to improve detection accuracy and reduce false positives.

CHAPTER 1

INTRODUCTION

1.1 Background Information

Ransomware is a type of malicious software that is designed to prevent users from accessing their files or computer systems by encrypting data and demanding a ransom for its recovery. In recent years, ransomware attacks have become a major cybersecurity concern for individuals, businesses, educational institutions, healthcare organizations, and government agencies. Attackers continuously develop new ransomware variants and techniques, making traditional security solutions less effective against unknown and rapidly evolving threats.

Ransomware commonly enters a system through phishing emails, malicious attachments, infected websites, compromised software, or vulnerable network services. Once it gains access, the malware may identify important files and rapidly encrypt them, making them unavailable to the legitimate user. Some advanced ransomware can also attempt to disable security software, delete backups, spread across connected systems, and steal sensitive information before encrypting files. These activities can result in data loss, financial losses, operational disruption, and privacy risks.

1.2 Project Objectives

The main objective of the Ransomware Detection and Prevention System is to develop an effective and proactive security solution for detecting and preventing ransomware attacks at an early stage. The system aims to continuously monitor system activities, file operations, and running processes to identify suspicious behavior such as rapid file modification, unauthorized file access, unusual file extensions, and abnormal encryption activities. When potential ransomware behavior is detected, the system should generate real-time alerts and take preventive actions such as blocking or terminating suspicious processes and restricting unauthorized file operations. Another important objective is to protect critical data through secure backup and recovery mechanisms, thereby reducing data loss and system downtime. The system also aims to provide a monitoring interface that displays detected threats, affected files, threat severity, and actions taken. Overall, the project focuses on providing reliable, automated, and behavior-based protection against both known and emerging ransomware threats while improving the security and availability of important data.

1.3 Significance of the Project

The Ransomware Detection and Prevention System is significant because ransomware attacks can cause serious data loss, financial damage, privacy issues, and disruption of important services. The proposed system provides a proactive approach to cybersecurity by continuously monitoring system activities and detecting suspicious ransomware behavior at an early stage. Unlike traditional security

solutions that mainly depend on known malware signatures, this system focuses on behavioral patterns such as rapid file encryption, unusual file modifications, suspicious processes, and unauthorized access. Early detection allows the system to generate alerts and take preventive actions before the attack causes extensive damage. The system also helps protect critical files through backup and recovery mechanisms, reducing the risk of permanent data loss. It can be useful for individuals, educational institutions, businesses, healthcare organizations, and other environments where protecting sensitive information is important. Overall, the project contributes to improving data security, reducing system downtime, minimizing the impact of ransomware attacks, and providing a reliable automated solution for modern cybersecurity challenges.

1.4 Scope of the Project

The scope of the Ransomware Detection and Prevention System is to develop a security application capable of monitoring computer system activities and identifying suspicious behavior associated with ransomware attacks. The system focuses on monitoring file operations, running processes, file modifications, file extensions, and other system activities that may indicate ransomware behavior. When suspicious activity is detected, the system can generate real-time alerts and perform preventive actions such as stopping malicious processes, restricting unauthorized file operations, and isolating affected resources. The project also includes mechanisms for protecting important files through secure backup and recovery. A monitoring interface can be provided to display detected threats, affected files, threat severity, and actions performed by the system. The system can be designed for use in personal computers, educational institutions, and small organizational environments. The scope can be further extended in the future by integrating machine learning-based detection, cloud-based monitoring, network-level ransomware detection, automated threat intelligence, and advanced incident response mechanisms.

1.5 Methodology Overview

The Ransomware Detection and Prevention System follows a systematic methodology consisting of data collection, system monitoring, behavior analysis, threat detection, prevention, alert generation, and recovery. Initially, the system continuously monitors file activities, running processes, system events, and other relevant activities. The collected information is then analyzed to identify abnormal patterns commonly associated with ransomware, such as rapid file modification, multiple file renaming operations, unusual file extensions, excessive file access, and suspicious process behavior. A detection mechanism evaluates these activities and determines whether the observed behavior is normal or potentially malicious. When ransomware-like activity is identified, the system generates an alert and assigns an appropriate threat level.

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Ransomware has become a major cybersecurity threat that can affect individuals, organizations, educational institutions, healthcare systems, and businesses. The primary problem is that ransomware can enter a computer system through phishing emails, malicious attachments, infected websites, vulnerable software, or unauthorized access. After gaining access, the ransomware may rapidly encrypt important files and make them inaccessible to the legitimate user. Attackers may then demand a ransom in exchange for restoring access to the encrypted data. This can result in significant data loss, financial damage, operational disruption, and privacy risks.

One of the major challenges in ransomware protection is the inability of traditional security mechanisms to detect new or modified ransomware variants. Many conventional antivirus systems depend heavily on known malware signatures.

2.2 Evidence of the Problem

Ransomware attacks have become a serious cybersecurity problem because they can quickly encrypt important files, disrupt computer systems, and cause financial and operational losses. Evidence of this problem can be observed through the increasing use of phishing emails, malicious attachments, vulnerable software, stolen credentials, and unauthorized access methods to distribute ransomware. Once ransomware enters a system, it can rapidly modify, rename, or encrypt a large number of files, making them inaccessible to users. Traditional antivirus systems that mainly depend on known malware signatures may not always detect newly developed or modified ransomware variants. In addition, attackers may attempt to disable security tools or compromise backups, making recovery more difficult. System activities such as sudden mass file modifications, unusual file extensions, excessive file access, suspicious process execution, and abnormal encryption behavior can serve as important indicators of a ransomware attack.

2.3 Stakeholders

The Ransomware Detection and Prevention System involves several stakeholders who benefit from the protection and security provided by the system. The primary stakeholders are individual users, organizations, educational institutions, system administrators, cybersecurity professionals, and management teams. Individual users benefit from protection against unauthorized file encryption and data loss. Organizations and educational institutions can use

the system to protect sensitive documents, databases, and other important resources from ransomware attacks. System administrators are responsible for monitoring alerts, analyzing suspicious activities, managing security controls, and taking appropriate preventive actions. Cybersecurity professionals can use the system to investigate detected threats and improve the organization's security measures. Management teams benefit from reduced financial losses, lower system downtime, and improved data security. Backup and recovery personnel may also be involved in maintaining secure copies of critical data and restoring systems when necessary.

2.4 Supporting Data and Research

Ransomware has become a significant cybersecurity concern due to its ability to rapidly encrypt files, disrupt services, and cause financial and operational losses. Research in cybersecurity shows that modern ransomware attacks are becoming more sophisticated, with attackers using phishing, malicious files, stolen credentials, vulnerable applications, and remote access services to gain entry into systems. Security research also indicates that ransomware can perform large-scale file modifications within a short period, making early detection important for limiting damage. Traditional signature-based antivirus mechanisms may have limitations when dealing with new or modified ransomware variants because previously unknown threats may not have an available signature. Therefore, recent research increasingly focuses on **behavior-based detection**, where activities such as abnormal file access, rapid file changes, suspicious process execution, unusual file extensions, and large-scale encryption are analyzed to identify potential attacks.

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The development and design process of the **Ransomware Detection and Prevention System** follows a systematic approach that includes requirement analysis, system design, implementation, testing, and deployment. Initially, the requirements of the system are identified, including real-time monitoring of file activities, process monitoring, ransomware behavior detection, alert generation, prevention, logging, and data recovery. Based on these requirements, the system is designed with different functional components such as file monitoring, process monitoring, behavior analysis, ransomware detection, alert generation, prevention, and backup and recovery. The file monitoring component observes activities such as file creation, modification, deletion, renaming, and changes in file extensions, while the process monitoring component identifies suspicious processes and their activities. The collected information is analyzed to detect ransomware-like behavior, including rapid modification of multiple files, unusual encryption patterns, excessive file access, and suspicious process execution. When the system identifies potentially malicious behavior, it generates a security alert and activates preventive measures such as terminating suspicious processes, restricting unauthorized file operations, or isolating affected resources.

3.2 Tools and Technologies Used

The following tools and technologies were utilized in the development of the **Ransomware Detection and Prevention System**:

- **Programming Language:** Python
- **File Monitoring:** Watchdog
- **Process Monitoring:** Psutil
- **Detection Technique:** Behavior-Based Ransomware Detection
- **Data Storage:** SQLite Database / CSV Files
- **Web Interface:** Flask / Streamlit
- **Development Environment:** Visual Studio Code (VS Code)
- **Libraries:** Pandas, NumPy, Watchdog, Psutil
- **Operating System:** Windows/Linux
- **Version Control:** Git (Optional)
- **Backup and Recovery:** Local Secure Backup
- **Logging:** Python Logging Module

These tools and technologies provide efficient system monitoring, file and process analysis, ransomware behavior detection, real-time alert generation, threat prevention, logging, and data recovery capabilities.

3.3 Solution Overview

The proposed Ransomware Detection and Prevention System is designed to provide a proactive solution for identifying and preventing ransomware attacks before they cause significant damage to the system. The system continuously monitors file activities, running processes, and system events to identify suspicious behavior associated with ransomware. Activities such as rapid modification of multiple files, unusual file extensions, excessive file access, suspicious process execution, and abnormal encryption patterns are analyzed by the detection module. When potentially malicious behavior is identified, the system generates a real-time alert and assigns an appropriate threat level. The prevention module then takes suitable actions, such as terminating the suspicious process, restricting unauthorized file operations, or isolating the affected system. Important files can be protected through secure backup and recovery mechanisms, allowing users to restore data after an attack. All detected activities, alerts, and prevention actions are recorded in a log for further analysis. A monitoring dashboard can also provide users or administrators with information about system status, detected threats, affected files, and actions taken.

3.4 Engineering Standards Applied

The project follows several recognized engineering, software quality, and cybersecurity standards:

IEEE Standards

- **IEEE 830** – Used for preparing the **Software Requirements Specification (SRS)** and documenting the functional and non-functional requirements of the ransomware detection system.
- **IEEE 1016** – Used for **Software Design Description (SDD)**, including system architecture, modules, data flow, and overall system design.

ISO Standards

- **ISO/IEC 25010** – Used as a software quality model focusing on **functionality, reliability, usability, performance efficiency, maintainability, and security**.
- **ISO/IEC 12207** – Used to guide the **software development life cycle**, including requirements analysis, design, implementation, testing, deployment, and maintenance.

Cybersecurity Standards

- **ISO/IEC 27001** – Considered for implementing **information security management practices** and protecting the confidentiality, integrity, and availability of system data.
- **OWASP Security Practices** – Followed to improve application security and reduce common vulnerabilities during system development.

Data Management and Security Practices

- **Data Validation and Logging** – Standard data validation, error handling, and logging techniques are followed to maintain accurate and reliable security records.
- **Backup and Recovery Practices** – Secure backup procedures are incorporated to support data recovery and minimize data loss during ransomware attacks.

These standards and practices help maintain **software quality, security, documentation consistency, reliability, data integrity, and maintainability** throughout the development of the Ransomware Detection and Prevention System.

3.5 Solution Justification

The proposed **Ransomware Detection and Prevention System** is justified by the increasing threat of ransomware attacks and the limitations of traditional security solutions in detecting new and modified ransomware variants. Signature-based antivirus systems mainly depend on previously identified malware patterns, which may not be sufficient against unknown threats. Therefore, the proposed system uses continuous monitoring and behavior-based analysis to identify suspicious activities such as rapid file modification, unusual file extensions, excessive file access, suspicious process execution, and abnormal encryption behavior. Early detection allows the system to generate alerts and take preventive actions before extensive damage occurs. The inclusion of process monitoring, real-time alerts, logging, and backup and recovery mechanisms further improves the overall security and reliability of the system. The solution is also practical because it can be implemented using commonly available technologies such as Python, Watchdog, Psutil, Flask or Streamlit, and SQLite or CSV storage. By combining detection, prevention, monitoring, and recovery in a single system, the proposed solution can reduce data loss, minimize system downtime, and provide an effective approach for protecting important files against ransomware attacks.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The developed **Ransomware Detection and Prevention System** was evaluated to determine its effectiveness in identifying and preventing suspicious ransomware-like activities. The system was tested under normal system conditions as well as controlled scenarios involving suspicious file operations and process activities. During testing, the system successfully monitored file creation, modification, deletion, and renaming activities and identified abnormal patterns such as rapid modification of multiple files and unusual file extensions. When suspicious behavior was detected, the system generated appropriate alerts and initiated the configured prevention mechanisms. The logging module successfully recorded detected activities, affected files, threat levels, and prevention actions for further analysis. The backup and recovery mechanism was also evaluated to verify that protected files could be restored when required. The results indicate that the proposed system can provide early detection and response to ransomware-like behavior, thereby reducing the possibility of extensive data loss and system disruption. Overall, the evaluation demonstrates that continuous monitoring, behavioral analysis, real-time alerting, and preventive actions can improve the security of computer systems against ransomware attacks.

4.2 Challenges Encountered

During the development of the **Ransomware Detection and Prevention System**, several challenges were encountered. One of the major challenges was identifying ransomware accurately without affecting normal file operations, as legitimate applications may also perform large numbers of file modifications. Designing suitable behavioral rules and detection thresholds was therefore important to reduce false alerts. Another challenge was continuously monitoring file and process activities without significantly affecting system performance. The system also had to distinguish between normal and suspicious processes while handling a large number of system events in real time. Implementing effective prevention mechanisms was another challenge because terminating or restricting processes incorrectly could affect legitimate applications and important system files. Maintaining secure backups and ensuring reliable file recovery also required careful implementation. In addition, testing ransomware-related behavior had to be performed in a controlled and safe environment to avoid accidental data loss. Integrating the monitoring, detection, alert, prevention, logging, and recovery modules into a single application also required extensive testing and debugging.

4.3 Possible Improvements

The **Ransomware Detection and Prevention System** can be further improved by incorporating advanced technologies and additional security features. Machine learning and artificial intelligence techniques can be integrated to identify complex and previously unknown ransomware behavior more accurately and reduce false positive alerts. The system can also be enhanced with **real-time network monitoring** to detect suspicious communication between infected systems and external servers. A centralized cloud-based monitoring system could be introduced to allow administrators to monitor multiple computers from a single dashboard. Advanced threat intelligence can also be integrated to obtain information about newly identified ransomware variants and attack patterns. The backup mechanism can be improved by using encrypted, versioned, and isolated backups to prevent attackers from compromising backup files. Additional security features such as email or mobile notifications, automatic system isolation, user authentication, role-based access control, and detailed security reports can also be implemented.

4.4 Recommendations

It is recommended that the **Ransomware Detection and Prevention System** be continuously updated to handle new ransomware variants and emerging attack techniques. Organizations should implement regular and secure backups of critical data and maintain at least one isolated backup that cannot be easily accessed or modified by an infected system. Real-time file and process monitoring should be enabled to identify suspicious activities at an early stage. The system should also be enhanced with machine learning-based behavioral analysis to improve detection accuracy and reduce false positives. Regular security updates, operating system patches, antivirus updates, and strong access controls should be maintained to reduce potential entry points for attackers. Users should be educated about phishing emails, suspicious attachments, unsafe websites, and other common ransomware delivery methods. Periodic testing of the detection, prevention, and recovery mechanisms should be performed to ensure that the system works effectively during an actual security incident.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

Academic Knowledge

The development of the **Ransomware Detection and Prevention System** provided valuable knowledge and practical experience in the areas of cybersecurity, software development, system monitoring, and problem-solving. Through this project, I gained a better understanding of how ransomware attacks operate, how malicious activities can affect files and system resources, and why early detection is important for protecting sensitive data. I learned how to use **Python** for developing security-related applications and gained practical experience with file monitoring, process monitoring, event logging, and behavioral analysis. I also learned how to identify suspicious activities and implement suitable detection and prevention mechanisms. The project improved my understanding of software development practices, including requirement analysis, system design, implementation, testing, debugging, and documentation. Working with tools such as **Visual Studio Code, Watchdog, Psutil, Flask/Streamlit, and SQLite/CSV** helped me gain practical experience in integrating different technologies into a single application. The project also improved my problem-solving and analytical skills because several technical challenges had to be identified and resolved during development and testing. In addition, I developed a better understanding of the importance of data backup, recovery, security logging, and continuous system monitoring. Overall, this project strengthened my technical knowledge, cybersecurity awareness, programming skills, documentation abilities, and confidence in developing practical software solutions.

5.2 Challenges Encountered and Overcome

During the development of the **Ransomware Detection and Prevention System**, I encountered several technical and practical challenges. One of the major challenges was understanding how to differentiate between normal file activities and ransomware-like behavior, since legitimate applications can also modify or access multiple files. This challenge was addressed by studying ransomware behavior and defining suitable detection conditions based on suspicious activities such as rapid file modifications, unusual file extensions, and excessive file access. Another challenge was implementing continuous file and process monitoring without causing significant system performance issues. This was overcome by using appropriate monitoring libraries and organizing the application into separate modules. Handling false alerts was also challenging, so the detection conditions and thresholds were tested and adjusted using different scenarios. Integrating the detection, alert, prevention,

logging, and backup modules required careful testing and debugging to ensure that the components worked together correctly. Safe testing was another important challenge because ransomware-related experiments could potentially damage files, so controlled test scenarios and sample files were used instead of actual malicious ransomware.

5.3 Application of Engineering Standards

During the development of the **Ransomware Detection and Prevention System**, various engineering and software development standards were applied to ensure the quality, reliability, security, and maintainability of the project. **IEEE 830** principles were followed to identify and document the functional and non-functional requirements of the system, while **IEEE 1016** guidelines were considered for describing the system architecture, modules, and design. **ISO/IEC 25010** software quality characteristics were considered to improve functionality, reliability, usability, performance efficiency, security, and maintainability. **ISO/IEC 12207** software life cycle practices were followed throughout the stages of requirement analysis, design, implementation, testing, and maintenance. Security practices based on **ISO/IEC 27001** were considered to protect the confidentiality, integrity, and availability of important data. In addition, proper data validation, error handling, logging, backup, and recovery practices were implemented to improve system reliability. Applying these engineering standards helped maintain a structured development process and ensured that the proposed system was developed with appropriate quality and security considerations.

5.4 Insights into the Industry

The development of the **Ransomware Detection and Prevention System** provided valuable insights into the cybersecurity industry and its current challenges. Through this project, I understood that cybersecurity is not limited to detecting known malware but also requires continuous monitoring, behavioral analysis, threat prevention, and rapid incident response. I learned that organizations increasingly require security solutions that can identify new and evolving threats before they cause significant damage. The project also provided an understanding of how cybersecurity professionals use monitoring tools, security logs, automated alerts, endpoint protection, backup systems, and incident response techniques to protect organizational resources. I gained awareness of the importance of secure software development, regular system updates, access control, data protection, and employee security awareness in preventing ransomware incidents. The project also showed me that cybersecurity solutions must balance security and usability because excessive restrictions or false alerts can affect normal business activities.

5.5 Conclusion of Personal Development

The development of the **Ransomware Detection and Prevention System** contributed significantly to my technical and personal development. Through this project, I gained practical knowledge of cybersecurity concepts, ransomware behavior, file and process monitoring, threat detection, prevention techniques, logging, and data recovery. I improved my programming skills by working with Python and related tools and gained experience in designing, implementing, testing, and debugging a complete software solution. The challenges faced during the project helped me improve my problem-solving, analytical thinking, time management, and troubleshooting abilities. I also learned the importance of following engineering standards, maintaining proper documentation, testing systems carefully, and considering security throughout the development process. Working on this project increased my confidence in handling real-world technical problems and helped me understand how theoretical cybersecurity concepts can be applied to practical applications. Overall, the project has strengthened my interest in cybersecurity and provided a strong foundation for further learning and development in areas such as artificial intelligence, machine learning, ethical hacking, digital forensics, and advanced threat detection.

CHAPTER 6

CONCLUSION

The **Ransomware Detection and Prevention System** was developed to address the growing threat of ransomware attacks and the risks associated with unauthorized file encryption, data loss, and system disruption. The proposed system provides a proactive approach by continuously monitoring file activities, running processes, and system behavior to identify suspicious ransomware-related patterns. When abnormal activities such as rapid file modification, unusual file extensions, excessive file access, or suspicious process execution are detected, the system can generate alerts and initiate appropriate preventive actions.

The project demonstrates the importance of early detection and rapid response in reducing the impact of ransomware attacks. The integration of behavioral analysis, real-time monitoring, alert generation, prevention mechanisms, logging, and backup and recovery provides a comprehensive approach to protecting important data. The system also highlights the limitations of relying only on traditional signature-based detection and demonstrates the value of behavior-based security techniques for identifying emerging threats.

Throughout the development process, the project provided practical experience in programming, cybersecurity, system design, testing, debugging, and documentation. Although challenges such as false positives, continuous monitoring, and safe testing were encountered, they were addressed through systematic development and controlled testing. The system can be further enhanced using machine learning, artificial intelligence, threat intelligence, network monitoring, cloud-based security management, and advanced recovery mechanisms.

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APPENDICES

Appendix A: Code Snippets

```
/**  
  
 * @license  
  
 * SPDX-License-Identifier: Apache-2.0  
  
 */  
  
import express, { Request, Response, NextFunction } from 'express';  
  
import path from 'path';  
  
import { createServer as createViteServer } from 'vite';  
  
import { MovieNode, GraphEdge, GraphData } from './src/types';  
  
import { BASE_MOVIES, computeEdgesForNodes, generateLargeScaleNetwork } from  
 './server/dataset';  
  
import { AdjacencyGraph, runBfsWithFrames, runDfsWithFrames, runOptimizedTraversal }  
 from './server/traversal';  
  
async function startServer() {  
  
  const app = express();  
  
  const PORT = 3000;  
  
  // JSON parsing and size containment  
  
  app.use(express.json({ limit: '10mb' }));  
  
  // Centralized Server-Side Database Cache  
  
  let currentBaseNodes: MovieNode[] = [...BASE_MOVIES];
```

```

let standardGraph: GraphData = {

  nodes: currentBaseNodes,

  edges: computeEdgesForNodes(currentBaseNodes)

};


// Pre-compiled Scaled Graph for DAA Evaluation to avoid calculating on every API call

console.log('Pre-compiling 5000+ Scaled DAA Simulation Graph for immediate traversal
metrics...');

const scaledGraph: GraphData = generateLargeScaleNetwork(5200);

console.log(`Scaled graph loaded with ${scaledGraph.nodes.length} nodes and
${scaledGraph.edges.length} connections.`);


// Auxiliary API: Reset custom database additions

app.post('/api/db/reset', (req: Request, res: Response) => {

currentBaseNodes = [...BASE_MOVIES];

  standardGraph = {

    nodes: currentBaseNodes,

    edges: computeEdgesForNodes(currentBaseNodes)

  };

  return res.json({ success: true, message: 'Database reset to default 40+ curated nodes.' });

});


// Central API: Fetch graph structural maps

app.get('/api/movies', (req: Request, res: Response) => {

  try {

```

```

const isScaled = req.query.scale === '5000';

const graphToUse = isScaled ? scaledGraph : standardGraph;


// Unpack fields for optional search & filters

const query = (req.query.search as string || '').toLowerCase().trim();

const genre = req.query.genre as string || 'all';

const rating = parseFloat(req.query.rating as string || '0');

const yearFrom = parseInt(req.query.yearFrom as string || '1900');

const yearTo = parseInt(req.query.yearTo as string || '2030');


let filteredNodes = [...graphToUse.nodes];


// Perform selective pruning

if (query !== '') {

    filteredNodes = filteredNodes.filter(

        n =>

            n.title.toLowerCase().includes(query) ||

            n.director.toLowerCase().includes(query) ||

            n.actors.some(a => a.toLowerCase().includes(query)) ||

            (n.franchise && n.franchise.toLowerCase().includes(query))

    );

}


if (genre !== 'all') {

```

```

    filteredNodes = filteredNodes.filter(n => n.genre.includes(genre));
  }

  if (rating > 0) {
    filteredNodes = filteredNodes.filter(n => n.rating >= rating);
  }

  filteredNodes = filteredNodes.filter(n => n.year >= yearFrom && n.year <= yearTo);

  // Re-compile sub-edges so we do not send orphaned edges across network (optimizes
  Canvas rendering)

  const nodeIdsSet = new Set(filteredNodes.map(n => n.id));
  const filteredEdges = graphToUse.edges.filter(
    e => nodeIdsSet.has(e.source) && nodeIdsSet.has(e.target)
  );

  return res.json({
    success: true,
    nodes: filteredNodes,
    edges: filteredEdges,
    totalNodesInSystem: graphToUse.nodes.length,
    totalEdgesInSystem: graphToUse.edges.length
  });
} catch (err: any) {
  console.error('Centralized Graph Fetch Failure:', err);
}

```

```

return res.status(500).json({

  success: false,

  error: ' Centralized Graph Service Recovery: Fallback engaged.',

  details: err.message || 'Unknown network index state exception'

});

}

});

// Action API: Dynamically register custom movie node

app.post('/api/movies/custom', (req: Request, res: Response) => {

try {

  const { title, genre, rating, year, actors, director, franchise, description } = req.body;

  if (!title || !genre || !rating || !year || !director || !actors) {

    return res.status(400).json({

      success: false,

      error: 'Input Validation Violation',

      details: 'Fields (title, genre, rating, year, director, actors) are strictly required.'

    });

  }

  const cleanTitle = (title as string).trim();

  const cleanId = cleanTitle.toLowerCase().replace(/[^a-z0-9]/g, '_');

```

```

// Safeguard against duplicates

if (currentBaseNodes.some(n => n.id === cleanId)) {

  return res.status(499).json({

    success: false,

    error: 'Circular/Duplicate Record Blocked',

    details: `The movie title "${cleanTitle}" already maps to index key "${cleanId}"`.

  });

}

const genresList = Array.isArray(genre) ? genre : (genre as string).split(',').map(g =>
g.trim());

const actorsList = Array.isArray(actors) ? actors : (actors as string).split(',').map(a =>
a.trim());

const newMovie: MovieNode = {

  id: cleanId,

  title: cleanTitle,

  genre: genresList,

  rating: Math.min(10.0, Math.max(1.0, parseFloat(rating))),

  year: parseInt(year),

  actors: actorsList,

  director: (director as string).trim(),

  franchise: franchise ? (franchise as string).trim() : null,

  popularity: 75, // custom user additions start with standard popularity node weight

```

```
    posterColor: `linear-gradient(135deg, hsl(${Math.floor(Math.random() * 360)}, 60%, 20%), hsl(${Math.floor(Math.random() * 360)}, 40%, 10%))`,
```

```
    description: description ? (description as string).trim() : 'User custom generated connection hub link.',
```

```
    isCustom: true
```

```
};
```

```
currentBaseNodes.unshift(newMovie);
```

```
standardGraph = {
```

```
  nodes: currentBaseNodes,
```

```
  edges: computeEdgesForNodes(currentBaseNodes)
```

```
};
```

```
return res.json({
```

```
  success: true,
```

```
  message: `Successfully integrated node "${cleanTitle}" under schema index "${cleanId}".`,
```

```
  node: newMovie,
```

```
  totalNodes: standardGraph.nodes.length
```

```
});
```

```
} catch (err: any) {
```

```
  return res.status(500).json({
```

```
    success: false,
```

```
    error: 'Dynamic DB Register Critical Failure',
```

```
    details: err.message
```

```

    });

}

});

// Algorithm API: Run graph traversal step-by-step

app.get('/api/traverse', (req: Request, res: Response) => {

try {

    const isScaled = req.query.scale === '5000';

    const graphToUse = isScaled ? scaledGraph : standardGraph;

    const algorithm = req.query.algorithm as string || 'bfs';

    const startId = req.query.startId as string;

    const targetGenre = req.query.targetGenre as string || undefined;

    const maxSteps = parseInt(req.query.maxSteps as string || '150');

    const maxDepth = parseInt(req.query.maxDepth as string || '2');

    if (!startId) {

        return res.status(400).json({

            success: false,

            error: 'Missing Traversal Target Anchor',

            details: 'Parameter "startId" is mandatory for graph route mapping.'

        });

    }

}

```

```

const graphEngine = new AdjacencyGraph(graphToUse.nodes, graphToUse.edges);

const startIds = startId.split(',').map(s => s.trim()).filter(Boolean);

const validStartIds = startIds.filter(id => graphEngine.getNode(id));

if (validStartIds.length === 0) {

  return res.status(404).json({

    success: false,

    error: 'Target Anchor Node Not Found',

    details: `Movie lookup failed for index ids "${startId}".`

  });

}

let result;

if (algorithm === 'bfs') {

  result = runBfsWithFrames(graphEngine, startId, maxSteps, targetGenre);

} else if (algorithm === 'dfs') {

  result = runDfsWithFrames(graphEngine, startId, maxSteps, targetGenre);

} else if (algorithm === 'bfs_optimized') {

  result = runOptimizedTraversal(graphEngine, startId, 'BFS', maxDepth, targetGenre);

} else if (algorithm === 'dfs_optimized') {

  result = runOptimizedTraversal(graphEngine, startId, 'DFS', maxDepth, targetGenre);

} else {

  return res.status(400).json({

    success: false,

```

```

        error: 'Algorithm Selector Mismatch',

        details: `Support available for: bfs, dfs, bfs_optimized, dfs_optimized. Inputted:
"${algorithm}".`

    });

}

return res.json({

    success: true,

    algorithm,

    startId,

    frames: result.frames,

    metrics: result.metrics,

    recommendations: result.recommendations

});

} catch (err: any) {

    console.error('Traversal Pipeline Crash Recovery:', err);

    return res.status(500).json({

        success: false,

        error: 'Centralized Algorithmic Sandbox Exception',

        details: err.stack || err.message

    });

}

});

// Analytics API: Run multi-algorithm benchmarking

```

```

app.get('/api/compare', (req: Request, res: Response) => {

  try {

    const isScaled = req.query.scale === '5000';

    const graphToUse = isScaled ? scaledGraph : standardGraph;

    const startId = req.query.startId as string || graphToUse.nodes[0].id;

    const targetGenre = req.query.targetGenre as string || undefined;


    const graphEngine = new AdjacencyGraph(graphToUse.nodes, graphToUse.edges);

    const startIds = startId.split(',').map(s => s.trim()).filter(Boolean);

    const validStartIds = startIds.filter(id => graphEngine.getNode(id));

    if (validStartIds.length === 0) {

      return res.status(404).json({

        success: false,

        error: 'Root Benchmark Movie Exception',

        details: `Could not verify index keys "${startId}"`.

      });

    }


    // Run parallel computations to track real metrics

    const bfs_res = runBfsWithFrames(graphEngine, startId, 150, targetGenre);

    const dfs_res = runDfsWithFrames(graphEngine, startId, 150, targetGenre);

    const bfs_opt = runOptimizedTraversal(graphEngine, startId, 'BFS', 2, targetGenre);

    const dfs_opt = runOptimizedTraversal(graphEngine, startId, 'DFS', 2, targetGenre);

```

```

return res.json({
  success: true,
  startNode: graphEngine.getNode(validStartIds[0]),
  comparisons: {
    bfs: bfs_res.metrics,
    dfs: dfs_res.metrics,
    bfsOptimized: bfs_opt.metrics,
    dfsOptimized: dfs_opt.metrics
  }
});
} catch (err: any) {
  return res.status(500).json({
    success: false,
    error: 'Critical Analytics Execution Fault',
    details: err.message
  });
}
});

// Centralized Error Handling Middlewares for uncaught route slips
app.use((err: any, req: Request, res: Response, next: NextFunction) => {
  console.error('Uncaught Client Exception Cascade:', err);
  res.status(500).json({
    success: false,

```

```

    error: 'Centralized Containment Guard active: Handled uncaught exception.',
    details: err.message || 'Critical pipeline timeout exception.'
  });
});

// Vite static middleware mapping for UI SPA assets hydration
if (process.env.NODE_ENV !== 'production') {
  const vite = await createViteServer({
    server: { middlewareMode: true },
    appType: 'spa',
  });
  app.use(vite.middlewares);
} else {
  const distPath = path.join(process.cwd(), 'dist');
  app.use(express.static(distPath));
  app.get('*', (req, res) => {
    res.sendFile(path.join(distPath, 'index.html'));
  });
}

app.listen(PORT, '0.0.0.0', () => {
  console.log('=====');
  console.log(` CINEMATIC CONNECTION RUNNING ON PORT ${PORT}`);
  console.log(` Context Mode: ${process.env.NODE_ENV || 'development'}`);
});

```

```

    console.log('=====');
  });
}

// Global exception bindings to fully eliminate Node container crash loops
process.on('unhandledRejection', (reason, promise) => {
  console.error(' CENTRAL REJECTION INTERCEPTED:', promise, 'REASON:', reason);
});

process.on('uncaughtException', (error) => {
  console.error(' CENTRAL FAULT CONTAINED:', error);
});

startServer();

```



